

Psychophysical Haptic Measurement of Vertical Perception: Elucidating a Hand Sensory Bias

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Abstract—The primary sensory modality for probing spatial perception can vary among psychophysical paradigms. In the subjective visual vertical (SVV) task, the brain must account for the position of the eye within the orbit to generate an estimate of a visual line orientation, whereas in the subjective haptic vertical (SHV) task, the position of the hand is used to sense the orientation of a haptic bar. Here we investigated whether a hand sensory bias can affect SHV measurement. We measured SHV in 12 subjects (6 left-handed and 6 right-handed) with a forced-choice paradigm using their left and right hands separately. The SHV measurement was less accurate than the SVV measurements ($-0.6 \pm 0.7^\circ$) and it was biased in the direction of the hand used in the task but was not affected by handedness; SHV left hand $-6.8 \pm 2.1^\circ$ (left-handed $-7.9 \pm 3.6^\circ$, right-handed $-5.8 \pm 2.5^\circ$) and right hand $9.8 \pm 1.5^\circ$ (left-handed $7.4 \pm 2.2^\circ$, right-handed $12.3 \pm 1.8^\circ$). SHV measurement with the same hand was also affected by the haptic bar placement on the left or right side versus midline, showing a side effect (left vs midline $-2.0 \pm 1.3^\circ$, right vs midline $3.8 \pm 1.7^\circ$). Midline SHV measures using the left and right hands were different, confirming a laterality effect (left hand $-4.5 \pm 1.7^\circ$, right hand $6.4 \pm 2.0^\circ$). These results demonstrate a sensory bias in SHV measurement related to the effects of both hand-in-body (i.e., right vs left hand) and hand-in-space positions. Such modality-specific bias may result in disparity between SHV and SVV measurements, and therefore cannot be generalized to vertical or spatial perception. © 2021 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: vertical perception, SVV, SHV, haptic, spatial orientation.

INTRODUCTION

Perception of spatial orientation is a diverse sensory process that allows interaction with the environment in accordance with the current state of body position. In this process, the position of the body is encoded by peripheral sensors such as vestibular otoliths and proprioceptive receptors. Each of these sensory modalities, however, is inherently different in its respective frame of reference. Therefore, different types of sensory information must be integrated by the brain into a common frame of reference in order to maintain a

coherent estimate of orientation in space (Mittelstaedt, 1997; Van Beuzekom and Van Gisbergen, 2000; Tarnutzer et al., 2009a; Kheradmand and Winnick, 2017). Such sensory integration depends on the reliability of sensory signals that varies in different physical settings (De Vrijer et al., 2008; Clemens et al., 2011; Alberts et al., 2016).

When visual spatial context is removed, the brain must rely on other sensory signals that encode the position of the body to determine the position and orientation of external stimuli. In this context, the perceived direction of gravity is often used as a fundamental reference to quantify orientation in space and to study its underlying neural processes. This is the basis for using psychophysical tasks with either visual or haptic stimuli to measure perceived direction of gravity (i.e., vertical perception). In the subjective visual vertical (SVV) task, participants use a visual stimulus (e.g., a visual line) to report their vertical perception in the absence of visual spatial cues. Since visual inputs are

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Abbreviations: SVV, subjective visual vertical; SHV, subjective haptic vertical; VH, visual-haptic.

used in this task to report orientation in space, the brain must account for the position of the eye within the orbit to estimate the orientation of the visual stimulus (De Vrijer et al., 2009; Alberts et al., 2016; Kheradmand and Winnick, 2017). In the subjective haptic vertical (SHV) task, however, a haptic stimulus (e.g., a haptic bar) is used to report vertical perception in the absence of any visual spatial cue. In this task, proprioceptive and tactile information from the hand is primarily used to estimate the orientation of the haptic stimulus relative to perceived direction of gravity (Gentaz et al., 2008; Schuler et al., 2010; Tarnutzer et al., 2012c). Therefore, SVV and SHV tasks rely on different sensory information, and a disparity in their measurements may reflect modality-specific biases in how the brain estimates the position of the eye in SVV task versus the position of the hand in the SHV task. In line with such difference in the task-related sensory contributions, prior studies have shown disparity in SVV and SHV measurements (Guerraz et al., 2000; Lejeune et al., 2004; Bortolami et al., 2006; Tarnutzer et al., 2012a; Fraser et al., 2015; Kheradmand and Winnick, 2017). Therefore, it is important to examine the modality-specific bias related to each task for comparison of SHV and SVV results across studies.

In the present study, we addressed the modality-specific bias in SHV measurement and asked whether it can be related to using the hand in the haptic task (i.e., a hand bias). We hypothesized that a hand bias would be related to sensory inputs that are used by the brain to estimate the position of the hand and detect the orientation of the haptic stimulus. To address this plausible mechanism, we examined the effect of hand position within the body by comparing SHV measurements between the right and left hands, and with respect to an individual preferential use of one hand (i.e., handedness). In addition, we examined the effect of hand position in space by changing the placement of the haptic stimulus for SHV measurements. We also asked whether a hand bias in the SHV task can be related to how the haptic inputs are used by the brain to estimate vertical orientation, or whether it is related to using the hand in a haptic task independent of measuring vertical perception. To elucidate such sensory mechanisms and factor out motor influences from the SHV measurements, we implemented a forced-choice paradigm in which the orientation of the haptic stimulus is determined by touching it at each trial, as opposed to manually adjusting the stimulus to perceived vertical orientation. The adjustment of the stimulus by moving it can introduce a motor bias, resulting in a gradual shift of perceptual responses in consecutive trials (Tarnutzer et al., 2012b; Tarnutzer et al., 2012c). We used a similar forced-choice paradigm for SVV measurement and for comparison with the SHV results.

EXPERIMENTAL PROCEDURES

Participants

The experiments were approved by the Johns Hopkins institutional review board and informed written consent

was obtained from all participants. Twelve healthy volunteers (29 ± 14 years; 6 female) participated in the study. All participants were screened and reported to be in good health without history of visual, vestibular, neurologic, or psychiatric illness. Of the 12 subjects, six were left-handed (36 ± 17 years; 4 female) and six were right-handed (29 ± 14 years; 2 female). All subjects completed the Edinburgh Handedness Inventory and the Tapley-Bryden proficiency test prior to the experiment to quantify their handedness (Oldfield, 1971; Salmaso and Longoni, 1985; Tapley and Bryden, 1985). In the Edinburgh Handedness Inventory, subjects indicated which hand they preferred to use in a questionnaire. The Tapley-Bryden's circle-marking test was also used to verify handedness measurement (Tapley and Bryden, 1985).

Experimental procedures

Three different tasks were used to assess perception of spatial orientation: the SHV, SVV, and visual-haptic (VH) task. The SHV task measured perceived gravitational vertical using a haptic stimulus. The SVV task measured perceived gravitational vertical using a visual stimulus and served as a reference to compare with results from the SHV task. The VH task measured perceived orientation of the haptic stimulus relative to the visual stimulus to determine whether any bias in perceived orientation of the haptic stimulus is independent of perceiving gravity. All tasks (Fig. 1B–E) were done with subjects in complete darkness, to eliminate all visual cues other than the visual stimulus.

Subjective Haptic Vertical (SHV). A metal bar consisting of two pieces with parallel edges served as the haptic stimulus (Fig. 1A), and it was placed 30 cm in front of either the subject's hand, or the midsagittal plane. Each piece was 26 cm long and had an edge which was 1.5 mm wide and 1.4 cm thick. The edges of the two pieces had a gap of 7 mm in-between and provided a haptic cue to detect the orientation of the bar by touching it. The center of the bar was mounted on a stepper motor with a holding torque of 0.26 Nm and a step size of 1.8° (Stepperonline). The motor rotated the bar to specific orientations for each trial, and afterwards an auditory signal (a double beep at 1000 Hz) indicated the start of a trial. Participants were instructed to extend their forearm while resting their elbow on a flat armrest, and to only use their index finger and thumb to touch the two edges of the haptic bar with a pinch grip following the auditory signal. The center of the bar was adjusted for each subject so they could comfortably reach the center of bar with their hand. With the opposite hand, subjects reported whether the stimulus was tilted to the left or right of what they perceived as vertical, by clicking the left or right button of a computer mouse. As soon as the subject provided a response, the trial ended with another auditory signal (a single beep at 1000 Hz) indicating that the motor was starting to move for the next trial. Subjects were given 10 s for each trial to feel the stimulus and respond. If the subject

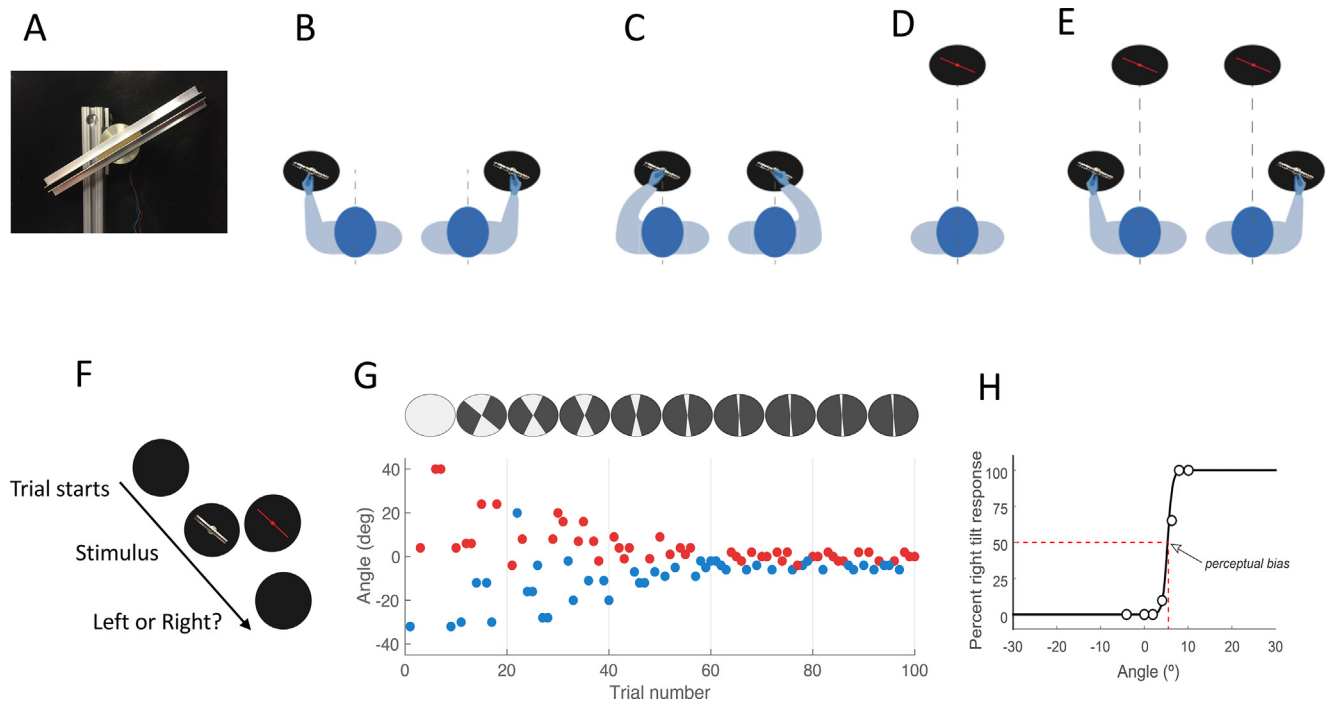


Fig. 1. The haptic stimulus in the SHV task was a bar that consisted of two pieces with parallel edges mounted on a stepper motor (A). The SHV task was done with using the left hand and right hand separately, and with placement of the haptic bar on the same side of the hand used (B) or at midline (C). In the SVV task (D), the orientation of a visual line was reported with respect to perceived vertical. In the VH task (E), the orientation of the haptic bar was reported relative to the orientation of the visual line. In the haptic tasks, the trial started with an auditory cue for subjects to touch the haptic bar (F). Sample time course of 100 trials with responses from the paradigm used in all three tasks (G). The y-axis shows the angle of the stimulus presented and the color indicates the response for that trial (left in blue and right in red). The stimulus angles were presented randomly within a range that started at 180° and then adjusted based on previous responses (gray sectors in the top circles). An example of psychometric fit to the responses in the paradigm (H). The bias value is determined as the center of the curve (i.e., point of subjective equality), where the probabilities of right and left responses are equal.

did not respond within 10 s, the trial was aborted, and a new trial started after a mouse button was clicked.

An MPU-9250 accelerometer sensor was attached to the haptic bar. Both the stepper motor and accelerometer sensor were connected to an Arduino Uno board and controlled by an Arduino-MATLAB interface (Arduino LLC, Italy). The stepper motor allowed the bar to rotate in the frontal plane relative to the subject. The accelerometer recorded the angle of the haptic stimulus, relative to gravitational vertical with a $\pm 0.1^\circ$ accuracy. The stepper motor moved to adjust the haptic stimulus with the target angle given by the paradigm. To ensure that the haptic bar did not move when it was touched during the trial, an accelerometer-to-software error of up to 1.2° was permitted for each trial. The trial was aborted for larger errors and the paradigm advanced to the next trial. In such cases, the missed trial was presented again later by the paradigm.

Subjective Visual Vertical (SVV). In each trial, a visual red line (length: 7.6° of visual angle, width: 0.13°) was presented around a red fixation dot (diameter: 0.33°), 135 cm away in front of the subject at the eye level. The task was to click a left or right button using a controller held with both hands, to report whether the upper half of the line was tilted to the left or right of perceived vertical. The fixation dot appeared first for 1 s, after which the visual line was presented for a minimum of

300 ms and maximum of 1.5 s until the subject responded. The trial was aborted, and the line disappeared if the subject did not respond within 1.5 s. Subsequently, a new trial started after a button was clicked in which the line was presented at a different angle. In such cases, the missed trial was presented again later by the paradigm.

Visual Haptic (VH). In each trial, subjects were presented with both visual and haptic stimuli simultaneously, and the task was to report the orientation of the haptic bar relative to the orientation of the visual line. The motor rotated the bar to a specific orientation in each trial, and afterwards an auditory signal (a double beep at 1000 Hz) indicated the start of a trial. After the double beep, subjects touched the bar with one hand (like the SHV task) and determined whether the haptic stimulus was tilted to the left or right relative to the visual line. The visual stimulus used in the VH task was the same as the one used for the SVV task, and responses were recorded by clicking left or right button on a computer mouse with the hand opposite to the one used for gripping the haptic stimulus. As soon as the subject provided a response, the trial ended with another auditory signal (a single beep at 1000 Hz) indicating that the motor was starting to move for the next trial. At each trial, subjects were presented with the visual and haptic stimuli for 10 s.

The trial was aborted if they did not respond within this time window and a new trial started after a button was clicked. Like the SHV task, an error in placement of the haptic bar less than 1.2° detected by the accelerometer was accepted by the paradigm. The trial was aborted for larger errors and the paradigm advanced to the next trial if there was such a mismatch error between the accelerometer reading and the target angle in the paradigm. In such cases, the missed trial was presented again later by the paradigm.

Adaptive two-alternative forced-choice (2AFC) paradigm

We used a custom paradigm written in MATLAB using Psychtoolbox (Kleiner et al., 2007) to present the stimuli and measure subject responses in the SHV, SVV, and VH tasks. The paradigm used for SHV and SVV was the same, while the paradigm used for VH task was a modified version. This adaptive paradigm was described in detail in previous studies, as depicted in Fig. 1F–H (Otero-Millan and Kheradmand, 2016; Winnick et al., 2019). For SVV and SHV tasks, the paradigm started with presenting the stimulus at random angles within the range of 180° , but as the paradigm continued, the range of angles was adjusted at every block of 10 trials. The new range was centered around the subjective vertical value calculated from the responses in previous 30 trials (for subjective vertical calculation see Data Analysis) and was decreased by half until the paradigm reached the 7th block, after which it remained constant at 8° for the remaining trials. Thus, the paradigm could adapt and track changes in subject's perception and was not biased by making a prior assumption about the subjective vertical values in the haptic and visual tasks.

In the paradigm used for VH task, the angle controlled adaptively was the relative angle between the visual line and the haptic bar. The paradigm started with random angles of the haptic bar within a range of 180° . The orientation of the visual line was selected randomly by the paradigm from a fixed set of angles (-8° , -4° , 0° , 4° , 8°). The range of angles was adjusted based on prior responses by decreasing in half for each block of 10 trials until the 7th block after which it remained constant at 8° for the remaining trials.

Experiment setup

In all tasks, subjects sat upright in a completely dark room with no windows, which was specially designed to perform experiments in the dark with all walls, floor, and ceiling in black color and doors sealed using thick drapes. For SVV and VH tasks, which required a visual stimulus, a CRT monitor (1280 by 1024 pixels) was placed 135 cm in front of the subject. To further eliminate any potential cues from the CRT monitor after subjects were dark-adapted, the line stimulus appeared in the center of a circular opening (diameter of 28 cm) of a black cardboard. The brightness and contrast levels of the screen was set to the minimum at a luminance of $<0.5 \text{ cd/m}^2$. The height of the CRT monitor was adjusted to align each participant's eye level with the

center of the monitor. Participants were instructed to wear their glasses if needed.

All participants completed the SVV, SHV and VH tasks in three separate sessions on separate days. The SVV task was performed in every session for comparison with the haptic tasks (Fig. 1D). The order of three sessions and the tasks during each session was counterbalanced among subjects. The haptic bar was placed on the right side of the subject for SHV measurement with the right hand and placed on the left side of the subject for SHV measurement with the left hand (Fig. 1B). The SHV was also measured in another session with the haptic stimulus at midline in front of the subject using the left and right hand separately (Fig. 1C). This was done to examine the effect of haptic bar placement (i.e., midline and left/right sides) versus the hand used in the SHV task (left/right hand). Subjects performed the VH task in a third session (again using the right and left hands separately) to examine whether a hand-induced bias only affected gravity perception (Fig. 1E).

In all tasks, subjects performed a set of 100 trials with the head immobilized in upright position using a molded bite bar. Across all tasks, a set of 100 trials lasted $4 \text{ min} \pm 6 \text{ s}$ (Mean $\pm$ Standard Error of the Mean (SEM)). Participants were monitored remotely through an infrared camera to ensure that they remained on the bite bar during the experiment and were following instructions during the task.

Data analysis

Using data collected from the 2AFC paradigm in each task, a psychometric curve was fitted to the trial responses using a cumulative gaussian function (MATLAB fitglm with probit link function) and a generalized linear regression model (Fig. 1H). With this method, the final SVV, SHV, and VH value were determined as the angle at which the probability of a left or right response was 50% in their corresponding tasks (point of subjective equality). The SVV, SHV, and VH precision were calculated as the difference between the 50% and 75% points on the psychometric curve per set of 100 consecutive trials.

The effect of hand use (i.e., hand bias) was determined in the haptic tasks (i.e., SHV and VH) by asking whether it could be due to placement of the haptic stimulus with respect to the subject (i.e., placement or side effect), or whether it could be related to a laterality in hand proprioception (i.e., laterality effect). The hand bias was calculated as the difference between the SHV with side placement of the haptic bar and SVV values for each subject. The side effect was calculated as the difference of lateral SHV and midline SHV using the same hand in each subject. The laterality effect was determined as the difference between the midline SHV and SVV measurements in each subject.

Statistical analyses were done in MATLAB using repeated measures analysis of variance (ANOVA) with the side and hand used as a within-subject factor, and handedness as a between-subject factor. To calculate correlations across subjects, we used Pearson method

to obtain the coefficient. A p -value less than 0.05 was considered significant.

RESULTS

Hand bias

The average measurement of vertical perception with the SHV and VH tasks are summarized in Table 1. The duration of an SHV or VH recording was on average $4.5 \text{ min} \pm 7 \text{ s}$ among the subjects. The duration of SVV recording was shorter with an average of $2.5 \text{ min} \pm 2 \text{ s}$. The average SVV in the three recording sessions were $-0.3 \pm 0.8^\circ$ (Mean $\pm$ SEM), $-0.6 \pm 0.7^\circ$, and $-0.8 \pm 0.6^\circ$, and there was no statistically significant difference among the three sessions (one-way repeated measures ANOVA, $F(2,22) = 0.43$, $p = 0.65$).

The average SHV was $-6.8 \pm 2.1^\circ$ using the left hand and $9.8 \pm 1.5^\circ$ using the right hand, with the haptic stimulus placed at the same side of the hand used in the task. The hand bias was calculated as the difference between the SHV and the SVV for each subject (Fig. 2). The left-hand bias was $-6.3 \pm 2.2^\circ$ and the right-hand bias was $10.6 \pm 1.4^\circ$. There was a significant hand bias towards the same side of the hand use with a rightward SHV bias using the right hand and a leftward SHV bias using the left hand (two-way repeated measures ANOVA, factor of hand: $F(1,11) = 51.3$, $p < 0.001$).

We examined the source of hand bias by asking whether it could be due to laterality in hand sensation (i.e., the laterality effect) or placement of the haptic stimulus (i.e., the side effect) (Fig. 2). The laterality effect using the left hand was $-4.5 \pm 1.7^\circ$ and $6.4 \pm 2.0^\circ$ using the right hand. The side effect was $-2.0 \pm 1.3^\circ$ using the left hand and $3.8 \pm 1.7^\circ$ using the right hand. Overall, when considering the hand bias, laterality effect, and side effect, there was a significant difference between the right and left hands in the SHV task (two-way repeated measures ANOVA; factor of hand: $F(1,11) = 35.17$, $p < 0.001$; hand bias, $p < 0.001$; laterality effect, $p = 0.007$; side effect, $p = 0.007$).

Hand bias specific to gravity perception?

To address whether the hand bias is specific to perception of gravity, VH task was used to measure perceived orientation of the visual line and haptic bar independent of gravity. The VH using the left hand was $-5.0 \pm 2.4^\circ$ and using the right hand was $10.6 \pm 1.9^\circ$. The SHV using the left hand was $-6.8 \pm 2.1^\circ$ and

using the right hand was $9.8 \pm 1.5^\circ$ (Table 1 and Fig. 3). The type of haptic task did not have a significant effect on the hand bias and there was no significant interaction between the hand use and the type of haptic tasks (two-way repeated measures ANOVA; factor of hand: $F(1,11) = 70.2$, $p < 0.001$, factor of task $F(1,11) = 1.23$, $p = 0.29$, interaction effect $F(1,11) = 0.24$, $p = 0.63$). The VH and SHV measurements were significantly correlated across subjects (Pearson $r^2 = 0.49$, $p < 0.001$) (Fig. 3). These findings show that the hand bias was present regardless of whether perceived vertical orientation or the orientation of the haptic bar relative to the visual line was reported.

Effect of handedness

The average Edinburgh handedness laterality quotient for the left-handers was -70 ± 23 (mean $\pm$ standard deviation (SD)), and 72 ± 23 for the right-handers, with negative numbers representing left-handedness and positive numbers representing right handedness. The range of laterality quotients of the participants was -37.5 to -90 for the left-handers and 44.4 to 100 for the right-handers. The average Tapley-Bryden laterality quotient for the left-handers was -0.17 ± 0.13 , and 0.28 ± 0.08 for the right-handers. The two laterality quotients were highly correlated across subjects ($r^2 = 0.92$).

The effect of handedness was examined in the haptic tasks. For the left-handed group, the SHV using the left hand was $-7.9 \pm 3.6^\circ$ and using the right hand was $7.4 \pm 2.2^\circ$. For the right-handed group, the SHV using the left hand was $-5.8 \pm 2.5^\circ$ and using the right hand was $12.3 \pm 1.8^\circ$ (Fig. 4). Overall, there was no significant effect of handedness on SHV or VH measurements (three-way repeated measures ANOVA, $F(1,10) = 1.84$, $p = 0.20$). No significant interaction was found between the handedness and the effect of hand use ($F(1,10) = 0.006$, $p = 0.94$).

Precision

As a measure of variability in SVV, SHV, and VH measurements, precision was calculated as the slope of the corresponding psychometric curves. The average precision of the left and right hands for SHV was $4.0 \pm 0.6^\circ$ and for VH was $4.3 \pm 0.4^\circ$ (Table 2). The average SVV precision was $1.2 \pm 0.2^\circ$. In the SHV and VH tasks, the effects of handedness and hand use were not

Table 1. Average SHV and VH results with negative values as leftward bias and positive values as rightward bias. SHV measurement was done with placement of the haptic bar at the left side, right side, and midline. The hand bias was calculated as the difference between SHV and SVV for each subject. The laterality effect was calculated by taking the difference of midline SHV and SVV for each subject. The side effect was calculated as the difference of lateral SHV and midline SHV for each subject

Hand Used	SHV $\pm$ SEM (Haptic bar placement)	SHV Hand Bias $\pm$ SEM	SHV Laterality effect $\pm$ SEM	SHV Side effect $\pm$ SEM ^o	VH $\pm$ SEM ^o
Left Hand	-6.8 ± 2.1 (left) -4.8 ± 1.5 (midline)	-6.3 ± 2.2	-4.5 ± 1.7	-2.0 ± 1.3	-5.0 ± 2.4
Right Hand	9.8 ± 1.5 (right) 6.1 ± 2.1 (midline)	10.6 ± 1.4	6.4 ± 2.0	3.8 ± 1.7	10.6 ± 1.9

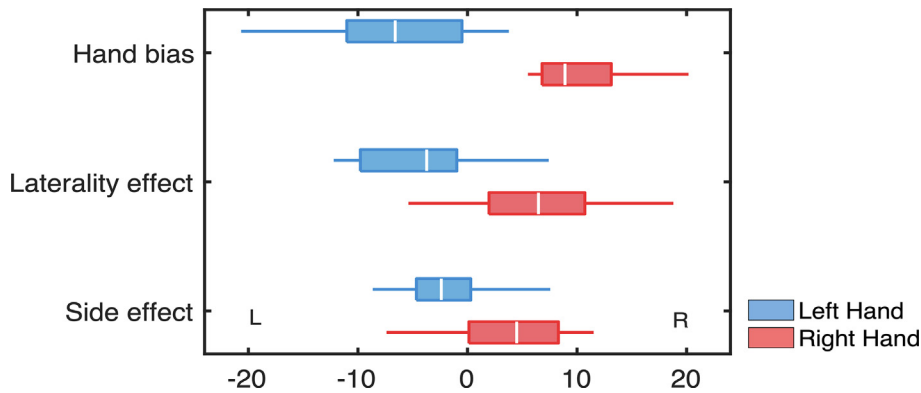


Fig. 2. The hand bias and effects of laterality and side placement of the haptic stimulus are shown for SHV task. The negative values indicate leftward bias (L), and the positive values indicate rightward bias (R). The box plots indicate the interquartile range with the median value (white line) and the whiskers indicate the 5th and 95th percentile range.

DISCUSSION

Direction of gravity is used as a fundamental frame of reference for quantifying orientation in space (Harris et al., 2014; Kheradmand and Winnick, 2017). However, the primary sensory modality for probing spatial perception can vary among psychophysical tasks. Our results show that the haptic vertical responses are affected by a hand bias. This bias was in the same direction of the hand use, with a leftward bias when SHV was measured with the left hand, and a rightward bias when it was mea-

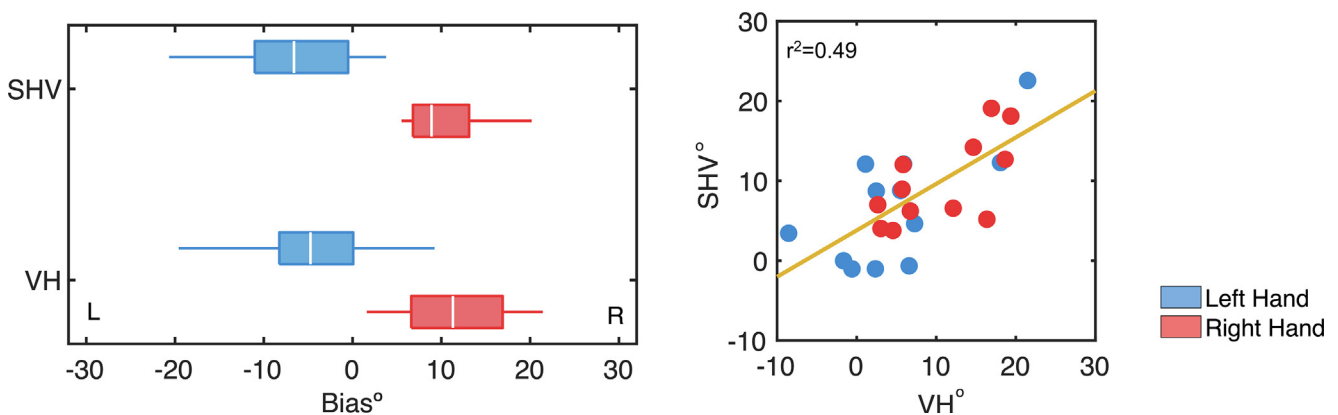


Fig. 3. Left: Both SHV and VH tasks show a bias towards the side of hand use (i.e., a rightward bias using the right hand and a leftward bias using the left hand). The box plots indicate the interquartile range with the median value (white line) and the whiskers indicate the 5th and 95th percentile range. Right: Correlation of SHV and VH tasks across subjects, ($r^2 = 0.49$, $p < 0.001$). SHV: subjective haptic vertical, VH: visual haptic.

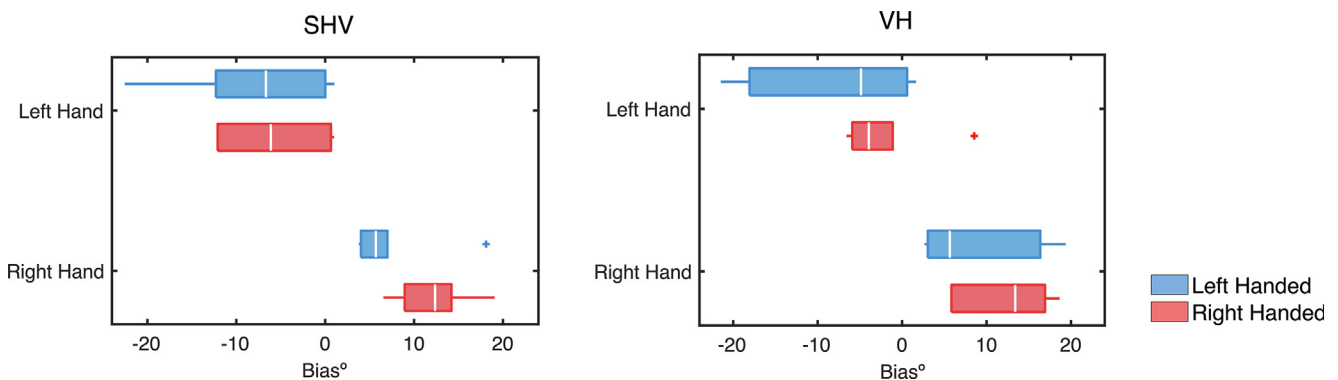


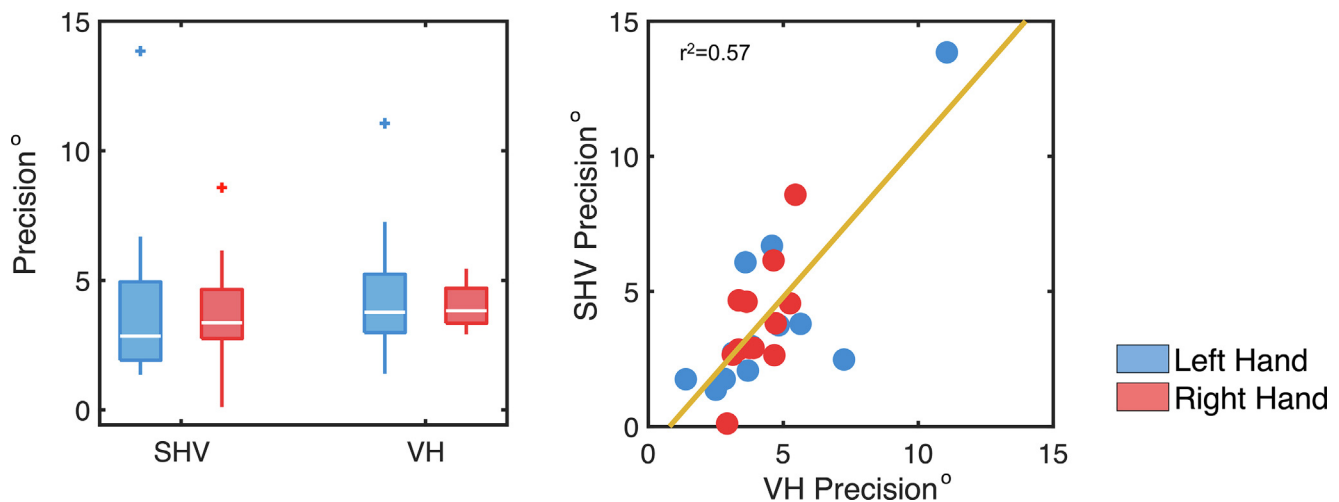
Fig. 4. SHV and VH results based on the hand use and handedness. The box plots indicate the interquartile range with the median value (white line) and the whiskers indicate the 5th and 95th percentile range. The + markers indicate outliers. SHV: Subjective haptic vertical, VH: Visual haptic.

significant (three-way ANOVA, factor of task: $F(1,10) = 0.48$, $p = 0.50$, factor of hand: $F(1,10) = 0.23$, $p = 0.64$, factor of handedness: $F(1,10) = 0.90$, $p = 0.37$). The SHV and VH precisions were significantly correlated across subjects ($r^2 = 0.57$, $p < 0.001$) (Fig. 5).

sured with the right hand. Since the motor bias was removed from the measurements, the hand bias can be attributed to sensory information used by the brain to estimate the position of the hand and detect the orientation of the haptic stimulus (i.e., tactile and proprioceptive inputs). In support of such sensory origin for the hand bias, there was no effect from handedness, and both left-handers

Table 2. SHV and VH precision based on the hand use and handedness

Hand Use	SHV $\pm$ SEM ^o		VH $\pm$ SEM ^o	
Left Hand	4.1 $\pm$ 1.0	3.2 $\pm$ 0.7 (Left-Handed) 5.0 $\pm$ 1.9 (Right-Handed)	4.5 $\pm$ 0.7	4.1 $\pm$ 0.8 (Left-Handed) 4.9 $\pm$ 1.3 (Right-Handed)
Right Hand	3.9 $\pm$ 0.6	3.1 $\pm$ 0.7 (Left-Handed) 4.7 $\pm$ 0.9 (Right-Handed)	4.1 $\pm$ 0.2	4.0 $\pm$ 0.4 (Left-Handed) 4.1 $\pm$ 0.3 (Right-Handed)
Overall	4.0 $\pm$ 0.6		4.3 $\pm$ 0.4	

**Fig. 5.** The SHV and VH precisions (left) and their correlations (right). The box plots indicate the interquartile range with the median value (white line) and the whiskers indicate the 5th and 95th percentile range. The + markers indicate outliers. SHV: Subjective haptic vertical, VH: Visual haptic.

and right-handers showed similar hand biases in the haptic tasks. SHV was also biased by the left- or right-side placement of the haptic bar, showing that in addition to the hand-in-body position (i.e., the laterality effect), the hand-in-space position (i.e., the side effect) also influenced haptic vertical responses. The hand bias was still present when the orientation of the haptic bar was reported relative to a visual line (i.e., the VH task), showing that the bias was primarily related to the haptic nature of the task and not confined to perception of gravity.

Prior studies have measured SHV only with manual adjustment of the haptic stimulus, and the effects of hand laterality and hand position in space were not investigated previously (Bauermeister et al., 1964; Schuler et al., 2010; Tarnutzer et al., 2012c). In these studies, the motor bias from manual adjustments resulted in a systematic shift in responses or hysteresis from moving the haptic stimulus (e.g., a rightward bias with rightward movement of the haptic bar). Also, with the manual adjustment, SHV measurements with the right hand were biased to the left side, while adjustments with the left hand were biased to the right side (Bauermeister et al., 1964). This bias from using one hand was removed when both hands were used together in the SHV task, suggesting mixed effects from manual adjustment and placement of the haptic bar (Tarnutzer et al., 2012c). There was no effect from handedness or grip of the haptic bar on the accuracy of SHV measurements (Tarnutzer et al., 2012c), but the precision of SHV responses was

affected by handedness. This finding also suggests a motor effect as left-handed participants showed lower precision with using their right hand when adjust the haptic bar to vertical orientation (Bauermeister et al., 1964).

In this study we use a novel force-choice task to remove the motor bias from haptic and visual vertical measurements. SVV measurements were accurate and remained within two degrees of earth vertical as shown also by previous studies (Van Beuzekom and Van Gisbergen, 2000; Tarnutzer et al., 2009a; Fraser et al., 2015; Kheradmand and Winnick, 2017). SHV measurements were, however, less accurate among the same subjects. Such discrepancy in SVV and SHV results can be attributed to different sensory possesses in each task. In the SVV task, the position of the eye in the orbit can affect perceived orientation of the visual stimulus. The SHV task, however, is not affected by the visual inputs or by the eye position. Instead, it relies on tactile and proprioceptive inputs from the hand to sense the orientation of a haptic bar and report vertical perception. Since all measurements were done in the upright position, the difference in SHV and SVV cannot be related to the position of the head and how it may affect perceptual responses in each task. When the head position is changed relative to gravity (i.e., when the head is tilted), a dissociation of sensory reference frames that encode the head (in space) and eye (in head) position can lead to systematic errors and reduced precision in SVV responses (Young et al., 1975; Bortolami et al., 2006; De Vrijer et al., 2008,

2009; Tarnutzer et al., 2009b; Barra et al., 2010; Kheradmand and Winnick, 2017; Unwalla et al., 2021). Like SVV, SHV responses become less accurate with head tilt, but the patterns of SHV and SVV errors have not been consistent across different studies (Guerraz et al., 2000; Lejeune et al., 2004; Bortolami et al., 2006; Tarnutzer et al., 2012a; Fraser et al., 2015; Kheradmand and Winnick, 2017). Our results suggest that such discrepancy could be caused by the effects of hand bias in the haptic task and measurement bias from different types of visual and haptic tasks used among various studies. Therefore, the difference in SVV and SHV measurements cannot be simply generalized to vertical or spatial perception, and future studies will have to control for these biases for accurate comparisons of SVV and SHV results.

The paradigm in this study employed random orientations of haptic and visual stimuli, and the range of angles presented was adjusted based on responses from previous trials. This ‘adaptive’ design removed a measurement bias from presenting the haptic or visual stimulus with a prefixed set of orientation angles by the paradigm (Mezey et al., 2004; Pagarkar et al., 2008; Toupet et al., 2015; Venhovens et al., 2016). Because we used various haptic and visual tasks, the recordings in individual subjects were divided into three separate sessions, and the order of recordings was also counterbalanced among the subjects to control for task-related fatigue. Also, the duration of a haptic or visual recording was on average about 4.5 min among the subjects, and neither of the tasks lasted long to cause fatigue. In addition, the precision of haptic recordings (i.e., SHV and VH) were comparable (Fig. 5), which suggests that fatigue did not affect any of these measurements. Finally, we found that placement of the haptic stimulus could bias SHV measurements in our paradigm, which can be attributed to the hand position in space in haptic vertical responses (i.e., the side bias). It was, however, not possible to determine whether the orientation of the haptic bar at each trial could affect perceptual responses. This is because of the adaptive design of the forced-choice task in this study, which included only few trials for some angles and did not allow to quantify the hand bias in relation to the orientation of the haptic bar. Future studies will have to design specific paradigms to address this question.

The present study contributes to an emerging literature suggesting that psychophysical measurements can offer insights into sensory processing for perception of spatial orientation. In this context, the task-specific constraints should be considered when the results are compared between SHV and SVV measurements. Our results show that in the absence of motor bias, a hand bias can affect SHV measurements. This bias can be related to how the hand-in-body (i.e., right vs left hand) and hand-in-space information is used by the brain to sense the orientation of the haptic stimulus.

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